

CE 511: Structural Dynamics with Application to Earthquake Engineering

Course Number : CE 511

Course Name: Structural Dynamics with Application to Earthquake Engineering

L-T-P-C: 3-0-0-3

Prerequisites: Strength of Materials and Structures (CE301) or Equivalent

Intended for: U G (3rd year and 4th year), PG

Distribution: Discipline Elective (UG); Core/Elective (PG)

Approval: 15th Senate

Course Contents

- **Introduction:** Overview and importance of structural dynamics; Various types of dynamic loading; Mass excited and base excited systems; Degrees of freedom (DOF); Typical force displacement behavior of structural elements/-systems. [3 Lectures]
- **Single Degree of Freedom (SDOF) Systems:** Dynamic equilibrium equation for SDOF system ; Analysis for undamped free vibration; Damping in structural system; Equation of motion and analysis for damped SDOF system ; Critically-damped, over-damped and under-damped systems; Estimation of damping by logarithmic decrement method; Classical and non -classical damping; Equivalent viscous damping; Response of undamped and damped SDOF system to harmonic excitation, Half-power band width method for damping evaluation; Response to support motion and force transmission to foundation; Response of SDOF system to periodic, impulsive, and general dynamic loading; Duhamel's integral. [13 Lectures]
- **Multi Degrees of freedom (MDOF) Systems:** Idealization of multi-story shear building as MDOF system; Equations of motion for two-story shear building; Natural frequencies of vibration , modes and mode shapes of MDOF system; Orthogonality of modes; Normalization of modes; Classically damped system; Static condensation of DOF for a multi-story building; Dynamic analysis of linear two-DOF system; Dynamic response control of structures, tuned mass damper. [10 Lectures]
- **Systems with Distributed Properties:** Vibration of uniform beam with various support conditions; Free vibration analyses; Natural frequencies of vibration and modes; Dynamic response to applied force and support excitation. [4 Lectures]
- **Application to Earthquake Engineering:** Equation of motion with earthquake excitation; Response quantities of interest and response histories; Response spectrum concept; Characteristics of acceleration, velocity and displacement response spectra ; Elastic design spectra; Modal analysis of MDOF systems; Modal superposition method and mode combination rules; Response spectrum analysis of multi-story building; Introduction to response history analysis; Indian Standard code provisions for computation of and analysis for earthquake loading; Introduction to seismic control of structures; Dynamics of base-isolated structures. [12 Lectures]

Textbooks:

1. A.K. Chopra, **Dynamics of Structures - Theory and Application to Earthquake Engineering**, 5th Edition, Pearson, 2017.
2. J.L. Humar, **Dynamics of Structures**, 3rd Edition, CRC Press, Taylor and Francis, 2012.



Reference Books:

1. E.L. Wilson, **Three-Dimensional Static and Dynamic Analysis of Structures**, 3rd Edition, Computers and Structures, 2002.
2. M. Paz, Leigh W., **Structural Dynamics - Theory and Computation**, 5th Edition, Springer Science+Business Media LLC, 2006.
3. R.W. Clough , J. Penzien, **Dynamics of Structures**, 3rd Edition, Computers and Structures, 1995.
4. T.K. Datta, **Seismic Analysis of Structures**, John Wiley & Sons (Asia) Pte Ltd, 2010.
5. IS 1 893 (Part I), **Criteria for Earthquake Resistant Design of Structures: Part I General Provisions and Buildings**, Bureau of Indian Standard, New Delhi , 2016.